

ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) incur prohibitive computational overhead when processing long videos, an overhead that traces directly to the very large number of visual tokens produced by their frame-level encoding. Prevailing efficiency methods curtail this cost by pruning or merging tokens according to importance or similarity, but in doing so they systematically neglect the dimension of video content most constitutive of its meaning—change and its turning points—and they possess no unified account of the coupled spatio-temporal relationships among tokens. We resolve both deficiencies through a single organizing principle: similarity identifies redundancy, whereas difference localizes the key events that carry a video’s narrative. Grounded in this principle, we introduce ST-SimDiff, a training-free token-selection framework that first assembles the visual tokens into a spatio-temporal graph, thereby subsuming their spatial and temporal associations within one unified representation. A parallel dual-selection strategy then operates over this graph: (1) similarity-based selection applies graph community detection to retain a compact set of representative tokens that condense static, redundant content, and (2) temporal-difference-based selection localizes content-changing points and preserves the tokens that encode the video’s decisive dynamic shifts. By coordinating these two complementary paths, ST-SimDiff faithfully represents both stable and dynamic content under a minimal token budget, and because it operates entirely on pre-computed features it applies directly to off-the-shelf MLLMs. Extensive experiments spanning three long-video understanding benchmarks and two architecturally distinct base models establish that ST-SimDiff consistently surpasses state-of-the-art compression approaches while sharply reducing inference time and memory, and that in several settings it matches or exceeds full-token accuracy. Our code will be released upon acceptance.

1 INTRODUCTION

The rapid maturation of Large Language Models (LLMs) has propelled a commensurate advance in Large Vision-Language Models (LVLMs), which now demonstrate remarkable competence across both image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). For video, contemporary LVLMs sample the input into a sequence of frames and encode each frame into hundreds—or even thousands—of visual tokens that the LLM subsequently consumes (Bai et al., 2025; Li et al., 2024). Effective as this paradigm is, its token count grows explosively with video duration and resolution; because the ensuing self-attention scales quadratically with sequence length, this growth imposes computational and memory burdens that sharply constrain the deployment of LVLMs in precisely the regimes where they are most valuable—long-video analysis and real-time interaction (Fu et al., 2024b).

A growing body of work confronts this bottleneck by improving LVLM efficiency, and these efforts organize into two principal families. The first is importance-based pruning. As FastV (Chen et al., 2024) observes, the attention distribution over visual information is markedly redundant across the layers of an LVLM, so that visual tokens receiving low attention scores in the deeper layers can be discarded to lighten the computational load. The second family is similarity-based merging and

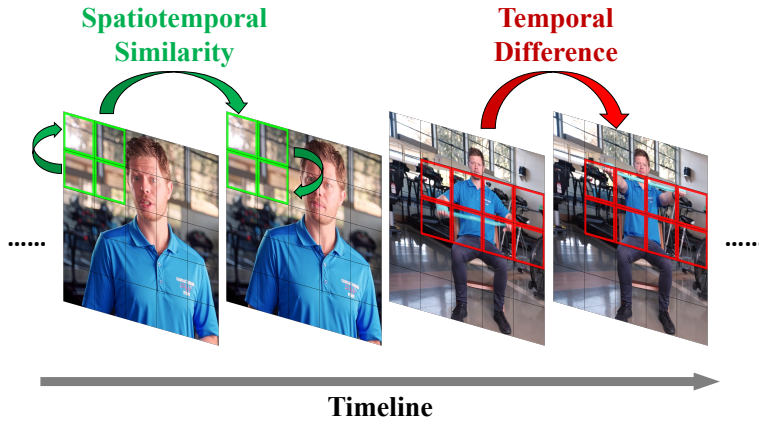


Figure 1: The core motivation of ST-SimDiff. Efficient video understanding must address two complementary regimes simultaneously: Spatiotemporal Similarity (left), exploited to compress information that is redundant both spatially within a frame and temporally across adjacent frames, and Temporal Difference (right), which must be detected to capture the key actions and events that drive the narrative. Existing methods attend to the former alone; ST-SimDiff treats the two on an equal footing, harvesting redundancy and preserving change within a single unified representation.

selection, which exploits the pronounced redundancy that visual tokens exhibit both across adjacent frames and within a single frame: FrameFusion (Fu et al., 2024b) reduces redundancy by merging similar tokens from neighboring frames, whereas VisionZip (Yang et al., 2025) directly selects dominant tokens at the vision-encoder level. Both families reduce the token budget effectively, yet each optimizes for a single facet of redundancy in isolation, and neither addresses the joint spatio-temporal structure that governs redundancy in video.

Despite their success at visual token reduction, existing methods are constrained in two decisive respects. First, they treat spatial correlations within a single frame and temporal correlations at fixed positions across frames in isolation; lacking any joint spatio-temporal model, they cannot characterize the coupled dynamic events that structure real video, and they consequently leave a substantial share of the redundancy inherent in video unharvested—a shortfall that becomes acute precisely under aggressive token reduction. Second, and more fundamentally, they share a common blind spot: by optimizing almost exclusively for informational commonality—similarity and importance—they neglect the changes and differences that carry a video’s meaning. Because a video’s narrative is driven by turning events, a method that pursues similarity alone smooths over a sudden action and, in turn, misreads the content. These two limitations are not incidental; they follow from a single underlying failure to model similarity and difference within one unified representation.

We resolve both limitations through a single organizing principle: similarity identifies redundancy, whereas difference localizes the key moments. On this view, an effective video token compression algorithm must satisfy two objectives jointly—representing a video’s stable content with the fewest possible tokens while precisely preserving its key changes. Guided by this principle, we propose **ST-SimDiff**, a dual token selection framework grounded in a spatio-temporal graph and composed of two complementary paths. The first is **similarity-based representative token selection**: treating every visual token as a node, we construct a graph over feature similarity in the joint spatio-temporal dimension, apply a graph community detection algorithm to recover tightly connected token clusters—each corresponding to a stable, persistent visual element—and retain from every cluster the few tokens of highest centrality as its representatives. The second is **difference-based event token selection**: we attend specifically to the edges that run along the temporal axis, and wherever the similarity between corresponding tokens of adjacent frames drops sharply, we designate a turning point and mark the associated tokens as critical event tokens for retention. The two token sets are finally merged and passed to the LLM, so that a single compact subset simultaneously encodes what persists and what changes. Because the entire procedure operates on pre-computed features without any additional training, ST-SimDiff applies directly to off-the-shelf LVLMs. Our contributions are summarized as follows:

- **A new principle for token compression.** We establish that token similarity and token difference warrant equal standing in VLM efficiency research, recasting difference from an overlooked artifact into a first-class selection signal and thereby closing the systematic blind spot of similarity- and importance-only methods.
- **A unified spatio-temporal framework.** We design a spatio-temporal graph that unifies the coupled spatial and temporal relationships among video tokens within a single representation, and upon it we build a training-free dual token selection strategy that concurrently screens for both representative tokens and pivotal event tokens; the procedure operates in time linear in the token count, adding only negligible overhead relative to the model’s inference cost.
- **Consistent empirical gains across models and benchmarks.** Through extensive experiments across three long-video understanding benchmarks and two architecturally distinct base models, we show that ST-SimDiff sharply reduces the visual token budget while consistently outperforming state-of-the-art compression methods and, in several settings, matching or surpassing full-token performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has substantially propelled multimodal understanding, giving rise to a family of prominent Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These models encode images or video frames into visual tokens through pre-trained visual encoders and forward the resulting tokens to an LLM, with an alignment module—commonly an MLP (Liu et al., 2024a;b) or a Q-Former (Li et al., 2023; Dai et al., 2023)—bridging the two modalities. This architecture endows LVLMs with exceptional capability across a broad range of multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and it is especially promising for video understanding, a setting that demands the processing of longer, higher-resolution, and structurally more complex inputs (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

This capability, however, comes at a pronounced computational cost that is most acute in the video setting. Because continuous sequences of frames must be encoded, the number of visual tokens grows steeply and far exceeds that of static images, straining both training and inference efficiency. Efficiently processing this large volume of video tokens while curbing computational overhead therefore remains a central open problem in LVLM research—and it is precisely the problem our framework targets.

2.2 VISUAL TOKEN COMPRESSION

Motivated by the substantial overhead of video processing, visual token compression has emerged as a central mechanism for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods approach compression along two principal axes: token importance and token similarity.

Earlier work concentrated on importance-based token pruning, which identifies and discards the visual tokens that contribute least to model performance, quantifying importance through attention scores or other feature statistics, as exemplified by FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). A subsequent line of work (Shang et al., 2024; Ye et al., 2025) additionally incorporates token similarity to suppress redundancy more effectively while preserving critical information, recognizing that even high-importance tokens may be mutually similar and hence informationally redundant. FrameFusion (Fu et al., 2024b), for instance, combines similarity-based merging with importance-based pruning; VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information; and VisPruner (Zhang et al., 2025b) exploits visual cues to remove redundant items.

Despite this progress on visual redundancy, existing work leaves two questions substantially undressed. First, it scarcely models the intricate, global spatio-temporal similarity relationships among

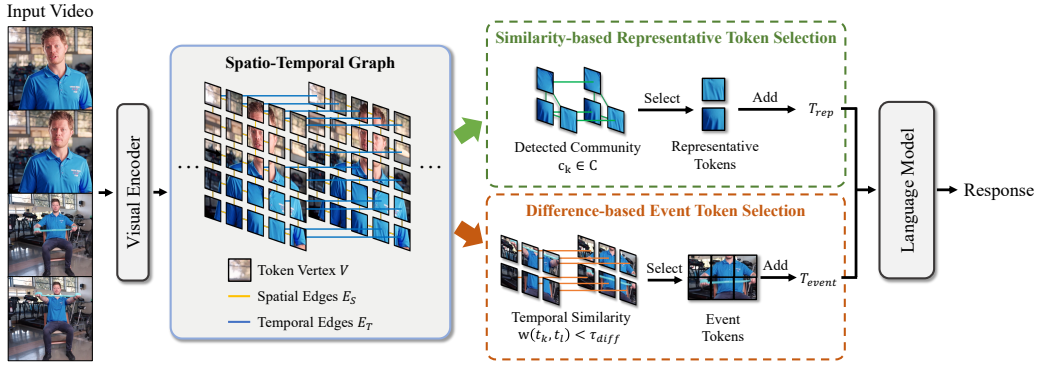


Figure 2: Overview of the ST-SimDiff framework. The pipeline proceeds through three stages: Spatio-Temporal Graph Construction, which encodes the coupled spatial and temporal token relationships within a single graph; Similarity-based Representative Token Selection, which compresses redundant, stable content via community detection; and Difference-based Event Token Selection, which localizes and preserves the tokens marking key dynamic transitions. The two selection paths operate in parallel and their outputs are merged, so that one compact subset simultaneously encodes what persists and what changes.

tokens: in video, visual information is redundant along the spatial and the temporal axis simultaneously, yet similarity-driven methods such as FrameFusion attend chiefly to the temporal similarity between adjacent frames and leave the coupled spatio-temporal structure unexploited—a gap that prevents them from fully harvesting the redundancy inherent in video and, in turn, curtails their performance under aggressive token reduction. Second, and more fundamentally, these methods optimize uniformly for commonality and thereby overlook difference—the turning points and abrupt events that carry a video’s narrative—so that the very content most diagnostic of “what happened” becomes the content most at risk of being pruned. ST-SimDiff closes both gaps at once: it models spatial and temporal similarity within a single graph and elevates difference to a first-class selection signal, and it is precisely in the aggressive-compression regime that this design yields its largest advantage.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video into a full sequence of N visual tokens, $T_{full} = \{t_1, t_2, \dots, t_N\}$. This encoding is computationally demanding: the subsequent self-attention mechanism scales quadratically with the token count N , incurring $O(N^2)$ complexity. Our objective is accordingly to design an efficient token selection function $f(\cdot)$ that maps the full sequence T_{full} , together with a target compression ratio r , to a substantially smaller subset $T_{sub} = f(T_{full}, r)$, so as to maximize the LVLM’s downstream task performance subject to the budget constraint $|T_{sub}| \approx r \cdot N$. The central challenge is thus to construct $f(\cdot)$ such that the retained subset T_{sub} concentrates the most critical information present in T_{full} —that is, such that it jointly preserves the video’s stable content and its defining changes under a stringent token budget.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff comprises two components that operate in parallel over a spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). The two components are complementary by design—one compresses what persists, the other preserves what changes—and in concert they realize the organizing principle that motivates the framework. The SRTS module identifies and condenses the video’s stable, redundant content: applying a commu-

nity detection algorithm to the graph, it groups highly similar tokens into clusters that correspond to persistent visual elements such as static backgrounds, and from each cluster it retains the few most central tokens to form the representative set T_{rep} . The DETS module, complementarily, captures the video’s crucial dynamic shifts: it analyzes the temporal edges of the graph and localizes the moments at which the similarity between corresponding tokens in adjacent frames drops sharply. Such points of high difference mark turning points, and the tokens framing these transitions are preserved as critical event tokens, forming the set T_{trans} . Combining the two paths, ST-SimDiff yields a final token subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ that is at once highly compact and information-rich, condensing stable content while precisely capturing the key events that define the video’s narrative.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens $T = \{t_1, t_2, \dots, t_N\}$, in which each token $t_k \in T$ occupies a distinct spatio-temporal location and carries a feature vector $\mathbf{x}_k$. We specify this location through three indices: the frame index $T(t_k)$, the spatial height index $H(t_k)$, and the spatial width index $W(t_k)$.

We encode the relationships among these tokens in a spatio-temporal graph $G = (V, E)$, whose vertex set V comprises all tokens in T and whose edge set decomposes as $E = E_S \cup E_T$, partitioning spatial and temporal relationships into two disjoint subsets so that each axis of redundancy can be addressed explicitly and independently. The spatial edges E_S connect tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

Complementarily, the temporal edges E_T connect tokens that occupy the same spatial position across adjacent frames, thereby encoding temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of every edge $w(v_i, v_j) \in E$ is defined as the cosine similarity between the feature vectors of its two incident tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

By construction, each token connects only to a small, constant number of spatial and temporal neighbors, rendering the graph sparse; it nonetheless encodes both local spatial relationships and temporal continuity within a single structure, furnishing the unified foundation on which our dual-path token selection strategy operates.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is inherently characterized by substantial spatio-temporal redundancy. In a static scene that persists for several seconds, the background tokens remain highly similar over time; likewise, the tokens tracking an object that moves across the frame retain strong semantic correlation despite their shifting spatial positions. Whereas existing methods treat temporal and spatial similarity in isolation, our spatio-temporal graph is expressly constructed to capture their joint structure: within it, semantically related tokens coalesce into dense “communities” or “clusters” by feature similarity, irrespective of their particular frame or spatial location. Retaining a small number of representatives from each community is therefore a principled and highly efficient strategy for compressing redundant information while preserving the core semantics of the scene.

To recover these strongly correlated clusters precisely, we first sparsify the graph G with a similarity threshold τ_{sim} , retaining only the edges whose weight exceeds it and thereby forming the pruned graph $G' = (V, E')$ with $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$; this thresholding step guarantees that a community is bound together by genuinely high feature similarity rather than by weak, incidental correlations. We then apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) to G' to recover the token clusters $C = \{c_1, c_2, \dots, c_m\}$. Within

each community $c_k \in C$, we rank and filter tokens by centrality, where the centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as its average similarity to all other tokens in the same community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

A higher centrality score thus identifies a token that is more representative of its community’s shared semantics. We then set the intra-community retention rate to the globally specified compression ratio r and preserve, from each community c_k , the top $\lceil |c_k| \cdot r \rceil$ tokens ranked by centrality. This uniform rule applies gracefully to every community, including the single-node communities formed by unique tokens, for which the sole node is necessarily retained. Formally, the representative token set T_{rep} is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score}=S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the k highest-scoring elements of a given set. In sum, by isolating and retaining the most central tokens within each semantic cluster, this strategy efficiently compresses the video’s static and persistent content while preserving its representative structure intact.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, difference defines its “events”. A model attuned solely to similarity may excel at recognizing “what is” yet falter on “what happened”, because a video’s plot is driven by key events whose essence is change—the appearance of a new object, the onset of an action, or a scene transition. In our formulation, such changes surface as a sharp drop in feature similarity between temporally adjacent tokens; accurately localizing these “turning points” of significant difference is therefore indispensable both for understanding a video’s dynamic content and for correctly answering “when” and “why” questions.

Difference along the temporal dimension is, in particular, a reliable indicator of a key event: the entry of a new object, a scene change, or the beginning or end of an action induces a pronounced change in the visual features at corresponding positions across adjacent frames. We accordingly analyze the temporal edges (E_T) of the spatio-temporal graph. For a temporal edge $(v_k, v_l) \in E_T$ joining two temporally adjacent tokens t_k and t_l , we compare the edge weight against a difference threshold τ_{diff} , which may be fixed or determined dynamically (e.g., as the 95th percentile of all temporal-edge difference scores); when the similarity falls below it, so that $w(t_k, t_l) < \tau_{\text{diff}}$, the transition is deemed significant and we retain the subsequent token t_l as a critical event token. Formally, the event token set T_{event} is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This strategy preserves precisely those tokens that mark moments of significant temporal change; by capturing these key transitions explicitly, it guarantees that the dynamic content central to the video’s narrative survives compression—content that similarity- and importance-driven methods are structurally prone to discard.

4.5 OVERALL REDUCTION PROCESS

The framework computes the representative set T_{rep} and the event set T_{event} in parallel and takes their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, as the initial candidate set. To meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$ exactly, we append a final pruning step: whenever $|T_{\text{candidate}}|$ exceeds N_{target} , we discard the $|T_{\text{candidate}}| - N_{\text{target}}$ least important tokens, where, following prior work (Chen et al., 2024), a token’s importance is read from its attention score in the shallow layers of the LLM. This step lets the method conform exactly to any prescribed computational budget while retaining the most critical information.

ST-SimDiff thus integrates two complementary mechanisms: content-driven graph-based token selection and attention-based dynamic pruning. The former exploits the intrinsic structure of the video content—its similarity and its difference—to guarantee that core information is retained, while the

latter supplies a flexible control that satisfies any target budget precisely. This coupling renders the compression process at once principled and adaptive, and it is what enables the framework to sustain a robust balance between efficiency and performance across compression regimes.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

We now analyze the computational overhead of ST-SimDiff, which decomposes across its three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS). We establish that every stage runs in time linear in the token count, so that the framework as a whole is linear.

Graph construction is the first stage. For each of the N visual tokens we compute the cosine similarity to its spatial and temporal neighbors; because every token has only a small, constant number of such neighbors, this reduces to a single pass over all tokens and costs $O(Nd)$, where d is the token feature dimension.

SRTS then identifies the token clusters. As detailed in our implementation (Section 5.2), we employ a connected-components algorithm whose near-linear complexity is $O(N + |E|)$; since $|E| \leq 3N$, this reduces to $O(N)$. The subsequent intra-community filtering could in principle become expensive—a naive all-pairs similarity computation within each community c_k would cost $O(|C| \cdot |c_k|^2 \cdot d)$ —so we cap the maximum community size, partitioning any community that exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$). This bound keeps the filtering step efficient and prevents it from dominating the overall cost. Although mainstream graph clustering algorithms such as the Louvain and Leiden methods admit richer community definitions, their higher complexity approaches $O(N \log N)$; we therefore adopt the simpler, highly efficient connected-components algorithm to minimize processing overhead, trading a marginally coarser community definition for a strictly linear cost.

DETS performs a single pass over all temporal edges, comparing each edge’s token similarity against the threshold to detect significant changes, again at a linear cost of $O(Nd)$.

Taken together, these linear-time operations govern the framework’s complexity and yield an overall cost of $O(Nd)$. This is markedly more efficient than the quadratic $O(N^2d)$ complexity of the LLM’s self-attention mechanism, and as a direct consequence ST-SimDiff delivers substantial token reduction while adding only a negligible computational footprint relative to the model’s primary inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate our method comprehensively, we adopt two architecturally distinct base models, LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c), and benchmark our approach against a representative spectrum of state-of-the-art video token compression techniques. These baselines span the field’s two dominant strategies: **importance-based reduction methods**, exemplified by FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods**, exemplified by VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). For a fair and complete comparison, we further bracket these methods with the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling of an equal number of tokens as a lower bound, so that every reported gain is situated precisely between these two reference points.

Benchmarks To probe our method across the breadth of long-video understanding, we evaluate on three challenging benchmarks that jointly span foundational understanding, fine-grained retrieval and reasoning, and egocentric causal inference: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME is a comprehensive benchmark for foundational video understanding, encompassing diverse video types across three length categories—Short, Medium, and Long—and thereby stressing performance over a wide range of temporal scales. LongVideoBench assesses the ability to retrieve and reason about fine-grained details in long videos through a “referring reasoning” task that spans four durations: 15s, 60s, 600s,

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video understanding benchmarks with LLaVA-Video-7B at 64 input frames (%), evaluated against importance-based and hybrid compression base-lines on VideoMME, LongVideoBench, and EgoSchema. For each metric, the best result among all compared methods is emphasized in **bold**. ST-SimDiff attains the top score in nearly every column at both the 30% and 50% token retain ratios; at $r = 50\%$ its VideoMME overall score (63.8) even surpasses the full-token upper bound (63.3), establishing that the method compresses without loss of accuracy—and, in this setting, with a marginal gain.

and 3600s. EgoSchema is a question-answering benchmark centered on long-term, first-person video that stresses the model’s grasp of character intentions and the causal chains linking actions. Taken together, the three benchmarks constitute a demanding and complementary testbed for the capabilities our framework targets.

5.2 IMPLEMENTATION DETAILS

All experiments run on NVIDIA L20 GPUs with 48 GB of VRAM under Ubuntu 22.04, and inference is evaluated with the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video we retain its original setting of 64 input frames, and to keep computational efficiency and resource usage directly comparable across backbones, we fix NVILA’s input frame count to 64 as well. We set the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$ —values whose robustness we establish in the threshold ablations of the appendix—and we report results at two token compression rates, $r = 30\%$ and $r = 50\%$. Balancing processing time against accuracy, we instantiate the community detection step with connected-component finding, which, as the complexity analysis shows, keeps the entire selection procedure linear in the token count.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

To evaluate ST-SimDiff comprehensively, we compare it against a range of mainstream efficient video language model (VLM) compression methods on three widely used long-form video understanding benchmarks—VideoMME, LongVideoBench, and EgoSchema. To establish the generalization and robustness of our method, we report detailed results on two architecturally distinct base

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video understanding benchmarks with NVILA-Video-8B at 64 input frames (%), under the same baselines, benchmarks, and retain ratios as in Table 1. For each metric, the best result among all compared methods is emphasized in **bold**. The gains observed on LLaVA-Video-7B transfer intact to this architecturally distinct base model—ST-SimDiff again leads every overall column at both retain ratios, and at $r = 50\%$ its VideoMME score (61.7) exceeds the full-token upper bound (61.5)—confirming that its effectiveness is architecture-agnostic rather than tied to a single backbone.

models, LLaVA-Video-7B and NVILA-8B, at token retain ratios of 30% and 50%. The results appear in Table 1 and Table 2.

(1) ST-SimDiff consistently outperforms every competing method across all test configurations—on both base models and at both retain ratios alike. Most tellingly, at the 50% token retain ratio its overall performance not only surpasses all other compression algorithms but, on several benchmarks, matches or exceeds the original model operating on the full token set: on LLaVA-Video-7B, ST-SimDiff reaches 63.8 on VideoMME against the 63.3 full-token upper bound, and on NVILA-8B it attains 61.7 against the 61.5 upper bound. That a 50% token reduction incurs no loss—indeed yields a slight gain—in accuracy directly substantiates the framework’s central claim, and it motivates a closer analysis of the baselines’ strategies and their limitations.

(2) Among the importance-based baselines (e.g., FastV, FasterVLM), FasterVLM is generally the stronger performer; their shared limitation, however, is an inefficient treatment of the temporal redundancy pervasive in video, as they tend to retain important yet repetitive tokens and thereby cap their compression efficiency. The hybrid baselines (e.g., PruMerge, VisionZip, FrameFusion) are more competitive than importance-only approaches, with FrameFusion typically the strongest among them. Even so, their common objective remains the identification and preservation of representative “commonality” information, which leaves a systematic blind spot: the key narrative information carried by “turning points” and “abrupt events” is liable to be overlooked—precisely the content that ST-SimDiff’s difference path is designed to retain.

(3) In sum, the baselines reflect a progressively deeper grasp of “redundancy” in video, yet two core challenges persist: first, the absence of a synergistic analysis of the joint spatio-temporal similarity among visual tokens, which impedes the capture of complex redundancy; and second, a general

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Ablation of the framework’s components on LLaVA-Video-7B (%), reported at the 30% and 50% token retain ratios. Starting from the importance-based Baseline, each added component yields a consistent gain: the similarity module (+ Sim)—and, within it, the spatio-temporal joint variant (Spa. + Tem.), which dominates its spatial-only and temporal-only counterparts—followed by the difference module (++ Diff). The complete framework attains the best score on every benchmark at both retain ratios, isolating the distinct and additive contribution of each design choice.

neglect of “difference” detection, which fails to actively preserve the key changes that define plot development. ST-SimDiff’s consistent gains follow directly from resolving both: it constructs a spatio-temporal graph that unifies these complex relationships and elevates difference detection to an explicit, first-class selection signal. These two design choices are thus not merely competitive on aggregate scores but address the very failure modes that limit prior work.

5.4 ABLATION STUDY

To isolate the contribution of each core component of ST-SimDiff, we conduct a detailed ablation study. Starting from a Baseline that performs only fundamental importance-based pruning, we progressively add the similarity selection module (+Sim) and then the difference selection module (++Diff), quantifying the effect of each in turn. All experiments are run at token retain ratios of 30% and 50%, with results reported in Table 5.

(1) We first assess the +Sim module. Introducing similarity modeling on top of the Baseline improves performance consistently across all three benchmarks. Decomposing the similarity strategy into spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.) variants establishes that jointly modeling the spatio-temporal associations among visual tokens is the most effective route to redundancy compression—the joint variant dominates both single-axis variants at each retain ratio—confirming that spatial and temporal similarity must be treated together rather than in isolation, exactly as our graph construction prescribes.

(2) Adding the difference selection module (++Diff) on top of the optimal spatio-temporal joint similarity module completes the framework and produces a marked performance gain that is most pronounced under the aggressive 30% compression ratio. This corroborates the module’s complementary role: whereas the +Sim module aggressively compresses redundancy, the ++Diff module acts as a safeguard that identifies and preserves the unique event tokens defining the video’s narrative. Its marginal gain narrows at the 50% ratio, where the more lenient +Sim module has typically already retained many of these event tokens and overall performance is approaching its upper bound. This dependence of the gain on the compression regime directly confirms our central motivation: the framework’s strength lies in balancing similarity-based redundancy compression against the essential preservation of difference-based events, and this balance proves decisive precisely in the demanding high-compression regime where the token budget is scarcest.

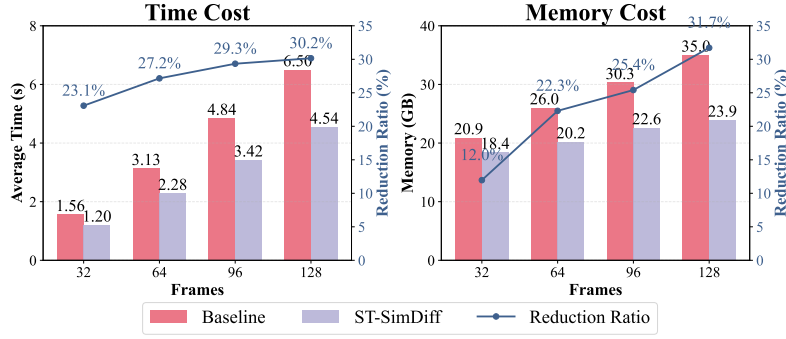


Figure 3: Computational cost of ST-SimDiff versus the uncompressed LLaVA-Video baseline, measured as average inference time and peak GPU memory usage under a 30% token budget across inputs of 32 to 128 frames. Both the time and the memory advantage widen monotonically as the frame count grows, so the benefit is largest precisely in the long-video regime that motivates the method—empirically corroborating the linear-cost advantage established by the complexity analysis.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the practical efficiency of ST-SimDiff, we measure its inference time and peak GPU memory usage under a 30% token budget and compare against the uncompressed baseline (Baseline LLaVA-Video). The results, reported in Figure 3 across input lengths from 32 to 128 frames, isolate the runtime and memory benefits of our compression at a fixed budget and empirically corroborate the linear-cost advantage established by the complexity analysis.

Inference Time ST-SimDiff delivers a substantial speedup that widens with the input length. As the input grows, the baseline’s average inference time climbs steeply from 1.56 seconds (32 frames) to 6.50 seconds (128 frames), whereas our method curtails this growth to only 4.54 seconds at 128 frames. Crucially, the relative time saved increases with video length, rising from 23.0% at 32 frames to 30.2% at 128 frames—precisely the regime in which efficiency matters most.

Memory Cost ST-SimDiff yields comparably strong gains in memory. The baseline’s peak GPU memory grows linearly with video length, from 20.9 GB (32 frames) to 35.0 GB (128 frames); by sharply reducing the number of tokens fed to the LLM, our method relieves this pressure, holding peak usage to 23.9 GB on 128-frame videos—a 31.7% saving over the baseline. This reduction renders it feasible to process longer videos on memory-constrained hardware, broadening the range of settings in which the model can be deployed.

Taken together, ST-SimDiff attains leading accuracy while simultaneously offering pronounced advantages in both inference time and memory, and both advantages scale with video length; the method therefore constitutes an efficient and practical solution for long-video understanding.

6 CONCLUSION

This paper confronts the computational inefficiency of Multimodal Large Language Models on long video—a problem rooted equally in the excessive number of visual tokens and in the systematic neglect of critical content change. We propose ST-SimDiff, a training-free framework that assembles the visual tokens into a spatio-temporal graph and models their coupled spatial and temporal associations within a single representation. On this graph a dual selection strategy operates in parallel: similarity-based selection identifies redundancy through community detection, while temporal-difference-based selection localizes and preserves the tokens at key transition points. Extensive experiments across three long-video benchmarks and two architecturally distinct base models demonstrate that ST-SimDiff substantially reduces both inference time and memory usage while consistently surpassing state-of-the-art compression methods and, in several settings, matching or exceeding full-token accuracy. More broadly, by elevating difference to equal standing with similar-

ity, this work establishes a new and principled direction for efficient video representation—one that holds the compression of stable content in balance against the preservation of the transitional events that carry a video’s meaning—and we anticipate that this similarity–difference perspective will inform token compression for video understanding well beyond the specific framework introduced here.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

To render the operation of ST-SimDiff concrete, Figure 4 presents representative samples of the intermediate and final token selections on a video sequence depicting a dynamic object-manipulation task. Consider first the Similarity-based Representative Token Selection (SRTS): the rows labeled “Cluster 1” and “Cluster 2” show the graph community detection algorithm at work, grouping spatially and temporally redundant regions—such as the static background and stationary objects—into cohesive communities. The visualization confirms that SRTS reliably isolates the stable visual elements and condenses them into a few central “representative tokens,” which appear as the sparse yellow boxes in the final results, so that the model absorbs massive background redundancy at minimal cost.

Complementing SRTS, the Difference-based Event Token Selection (DETS) proves decisive at the dynamic moments. In the “Final Results” row, the tokens triggered by high temporal difference are marked with dense red bounding boxes that coincide with the moving hand and the shifting blue liquid, confirming that our method precisely localizes the “turning points” at which the temporal difference crosses the threshold (τ_{diff}). By preserving this fine-grained action detail and rapid visual change, DETS guarantees that the model captures the essence of the event rather than being swamped by static content.

The complementarity of the two components is most evident in the overlay of the final results, where the balance is direct: the sparse yellow tokens represent the stable background while the dense red tokens capture the rapid motion. This contrast constitutes direct visual evidence for our core motivation—ST-SimDiff avoids the blind spot of similarity-only approaches by sampling densely through rapid change, while simultaneously pruning the spatiotemporal redundancy that importance-based methods routinely retain. The framework thereby strikes a favorable balance between computational efficiency and the preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison on long-form video understanding benchmarks with Qwen2.5-VL-7B at 64 and 128 input frames (%). At the 50% retain ratio, ST-SimDiff matches the full-token upper bound on VideoMME and closely approaches it on LongVideoBench, establishing that its effectiveness extends to a third, more recent model architecture and is in no way confined to the two backbones of the main experiments.

To further establish the generalization of our framework, we conduct additional experiments on the recently released Qwen2.5-VL model, evaluating on VideoMME and LongVideoBench to verify that the method’s efficacy is not tied to a particular model architecture but transfers across backbones.

The results in Table 4 attest to the strong performance of ST-SimDiff on Qwen2.5-VL. At a 30% token retention ratio, our method scores 62.4 on VideoMME and 56.8 on LongVideoBench; raising the budget to a 50% retention ratio lifts these to 62.9 and 58.4, respectively. Notably, at the 50% ratio the VideoMME score matches the upper bound obtained with the full token set, and the LongVideoBench score closely approaches its full-token counterpart. Together with the LLaVA-

Video and NVILA results of the main paper, these findings affirm that the framework generalizes robustly across a diverse set of model architectures.

C ABLATION STUDY RESULTS ON NVILA

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Ablation of the framework’s components on the NVILA-Video-8B model (%). The trends mirror those observed on LLaVA-Video-7B: the similarity module (+ Sim), its spatio-temporal joint variant (Spa. + Tem.), and the difference module (++ Diff) each contribute consistent gains, and the complete framework is best at both retain ratios—direct evidence that the component-level contributions established in the main paper are model-agnostic.

To establish that each component generalizes across base models, we replicate our detailed ablation on the NVILA-Video-8B model (Liu et al., 2024c). The setup follows that of the main-paper ablation exactly: starting from a Baseline that performs only fundamental importance-based pruning, we progressively add the similarity selection module (+ Sim) and then the difference selection module (++ Diff).

As Table 5 shows, the results reaffirm the framework’s design and track closely with the trends observed on LLaVA-Video-7B. Adding the similarity module (+ Sim) on top of the Baseline yields a steady improvement at both the 30% and 50% token retain ratios. Within this module we compare three variants—spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.)—and the joint variant proves the most effective: at $r = 50\%$, for instance, it reaches 61.4% on VideoMME, surpassing both the spatial-only and temporal-only strategies. This again underscores that redundant information is identified most effectively when spatio-temporal associations are modeled jointly rather than one axis at a time. Adding the difference selection module (++ Diff) on top of the optimal spatio-temporal joint similarity module completes the ST-SimDiff framework and, as the table makes clear, produces the largest gains. This outcome establishes the effectiveness and cross-model generalization of the framework’s two core modules—similarity and difference selection—and reconfirms, on a second architecture, our central idea of handling redundancy and change simultaneously.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

The similarity threshold τ_{sim} and the difference threshold τ_{diff} are the two key hyperparameters of our framework, governing the selection of representative and transitional tokens respectively. To characterize their effect, we ablate each in turn, with results reported in Figure 6. Both parameters induce the same qualitative trend—performance rises and then falls around an optimum—while exhibiting strong robustness across a broad surrounding range. The two failure modes are interpretable and symmetric: for τ_{sim} , too low a value yields imprecise community detection, whereas too high

a value fragments the integrity of the semantic clusters; for τ_{diff} , too low a value admits noise through over-sensitivity, whereas too high a value causes key events to be missed. The model attains its optimum at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, and we accordingly adopt this configuration throughout the remaining experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

To confirm that the robustness of our key hyperparameters holds on a second architecture, we repeat the threshold ablation on the NVILA-Video-8B model. As Fig. 6 shows, we evaluate the effect of the similarity threshold τ_{sim} and the difference threshold τ_{diff} on performance across VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . Sweeping τ_{sim} over $[0.6, 0.95]$, we find performance to be largely insensitive to the threshold. On VideoMME it peaks at 60.2% at $\tau_{sim} = 0.8$, and on LongVideoBench it is likewise optimal at $\tau_{sim} = 0.8$, reaching 55.2%. Although performance dips slightly when the threshold is set too high or too low, the overall fluctuation is minimal, establishing that the method remains stable across a wide range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . Sweeping τ_{diff} over $[0.1, 0.4]$, we again observe strong robustness. On both benchmarks the optimum is attained near $\tau_{diff} = 0.2$ —60.2% on VideoMME and 55.2% on LongVideoBench—indicating that a difference threshold around 0.2 most effectively captures key events and thereby maximizes performance under token compression.

In sum, these studies confirm the robustness of our method to its key hyperparameters and validate the choice of $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ in the main experiments: this single configuration consistently attains optimal, or near-optimal, performance across both benchmarks and both base models, so that no per-model tuning is required.

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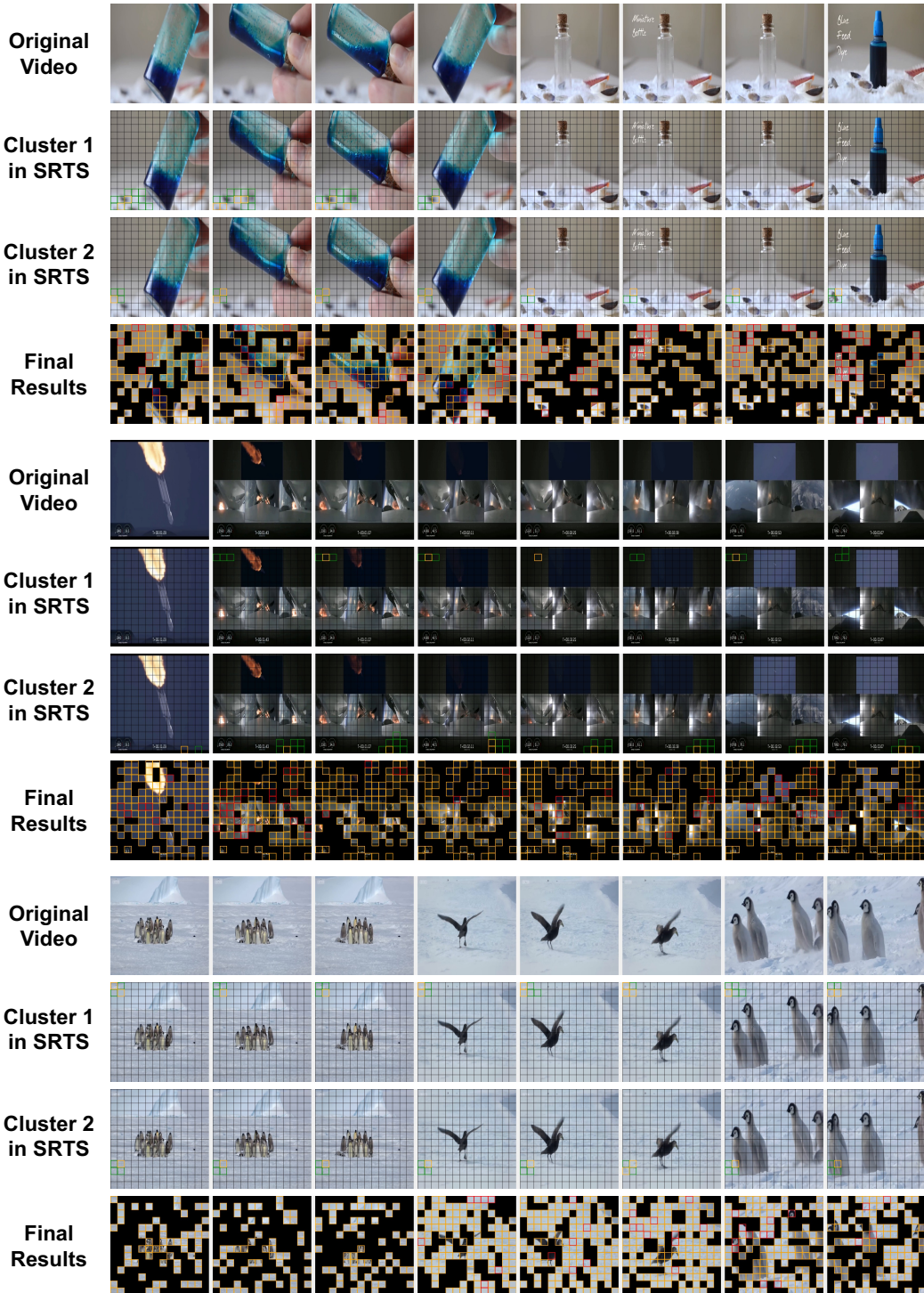


Figure 4: Visualization of the ST-SimDiff process. The token selection is decomposed into the communities recovered by SRTS (shown as grids in Cluster 1 and Cluster 2) and the event-driven tokens localized by DETS. The final result renders the complementary roles of the two modules explicit: sparse representative tokens (yellow) encode the stable content, whereas dense event tokens (red) capture the dynamic actions.

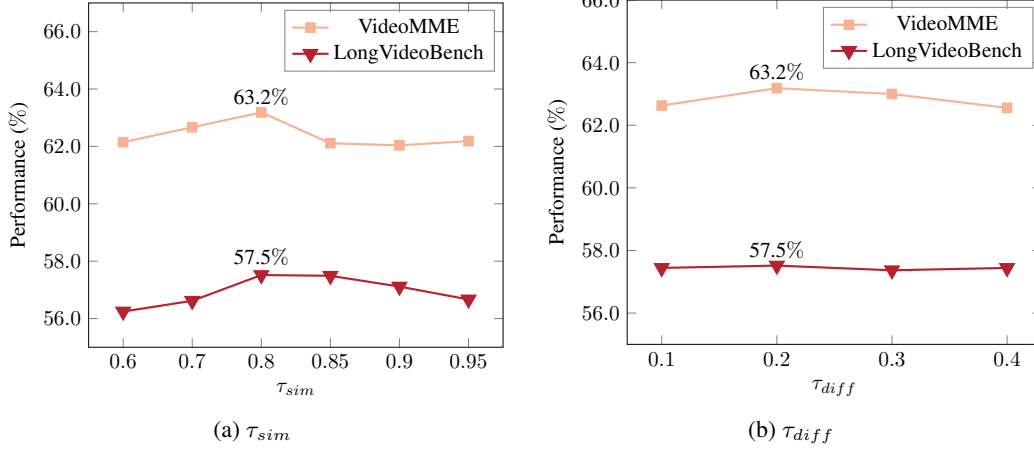


Figure 5: Ablation over the values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench (LLaVA-Video). Performance rises and then falls around a clear optimum for both thresholds while remaining stable across a broad surrounding range, demonstrating that the method is robust to the precise choice of either hyperparameter.

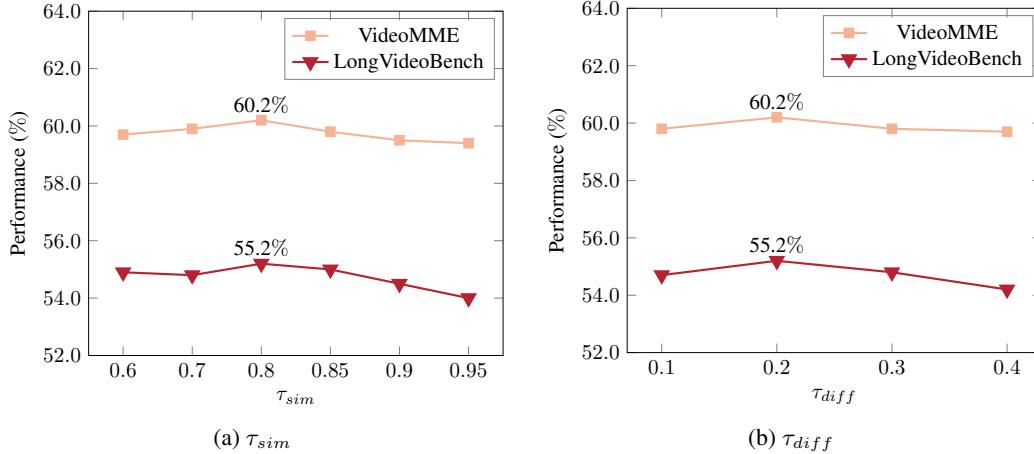


Figure 6: Ablation over the values of τ_{diff} and τ_{sim} on NVILA-Video-8B. As on LLaVA-Video, both thresholds peak at the same operating point ($\tau_{sim} = 0.8$, $\tau_{diff} = 0.2$) and vary only marginally across a wide range, direct evidence that the robustness of our hyperparameters is model-agnostic.